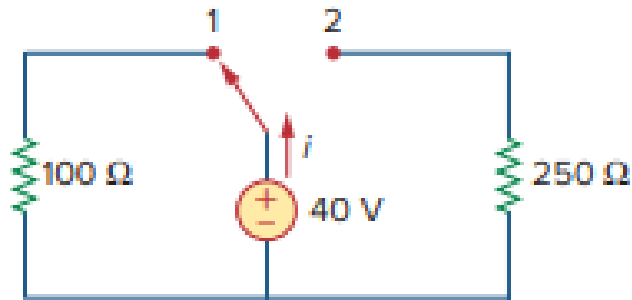


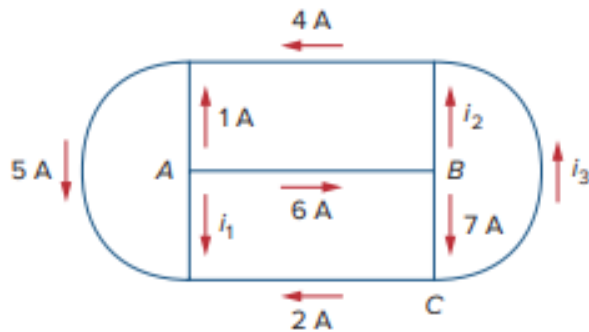
Nodes, Branches & Circuit Reduction

Tutorial

1. Find the current I when the switch is in position 1 and then when in position 2.

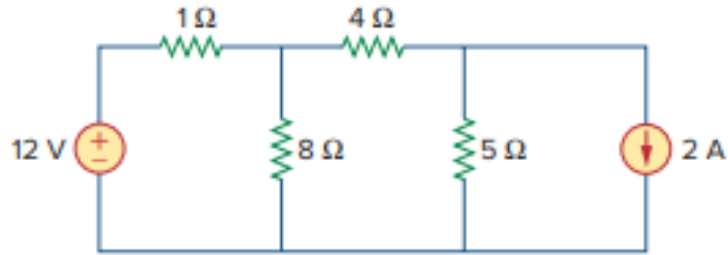


2. Find the currents i_1 , i_2 , i_3 .

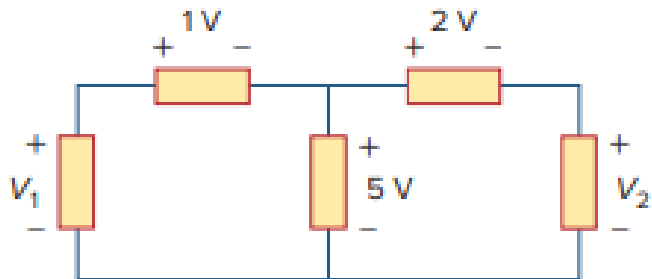


Tutorial

3. Determine the number of branches and nodes in the circuit given below.

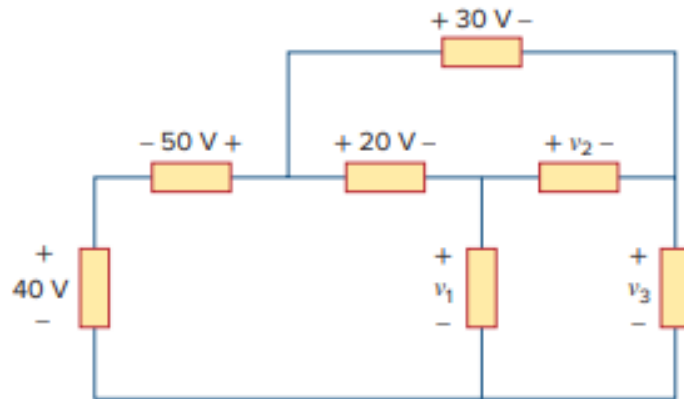


4. Calculate V_1 and V_2 .

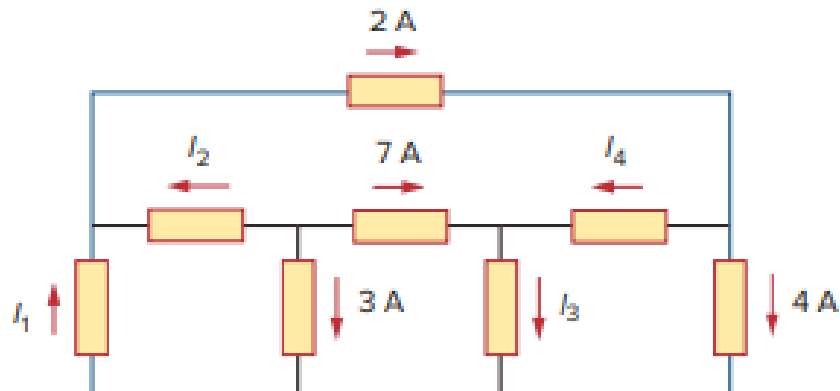


Tutorial

5. Calculate V_1 , V_2 and V_3

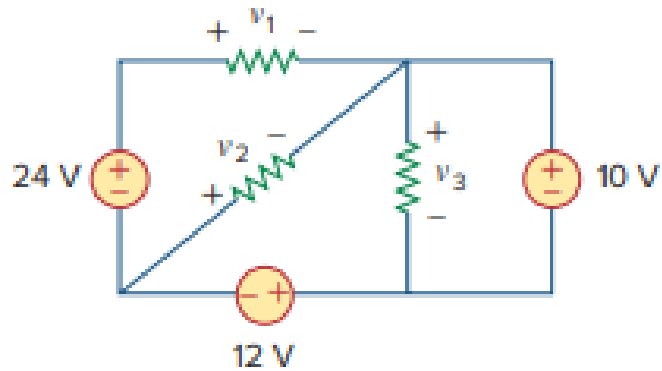


6. Find currents i_1 to i_4 .

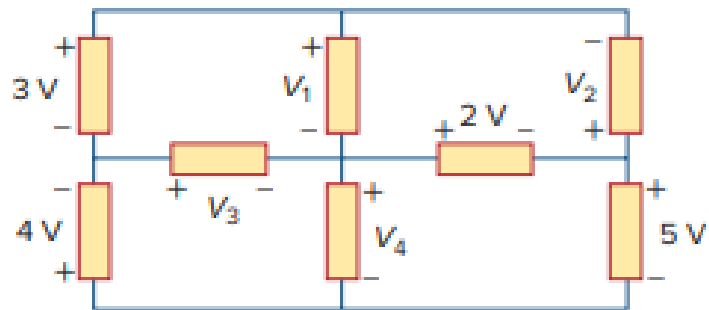


Tutorial

7. Determine V_1 through V_3 in the circuit.

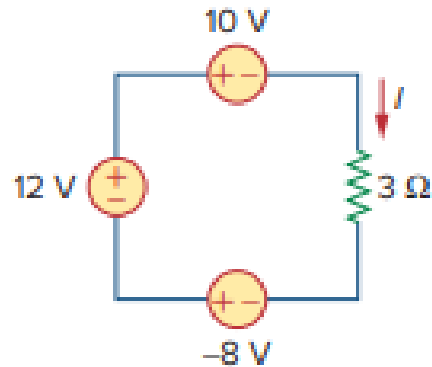


8. Calculate V_1 to V_4 using KVL.

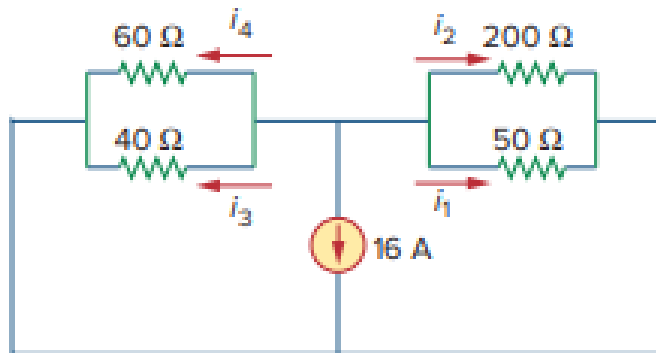


Tutorial

9. Find I , the power dissipated by the resistor, and the power supplied by each source.

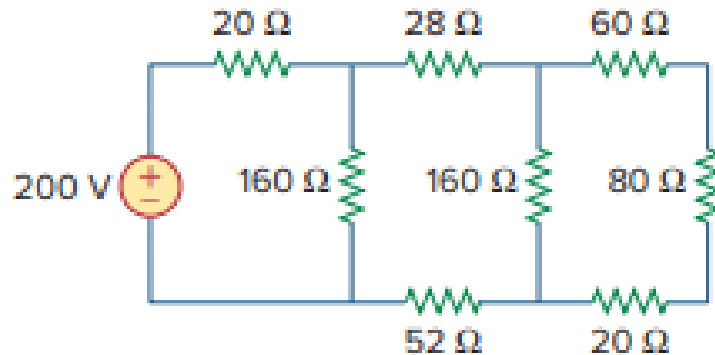


10. Calculate i_1 to i_4 in the circuit.

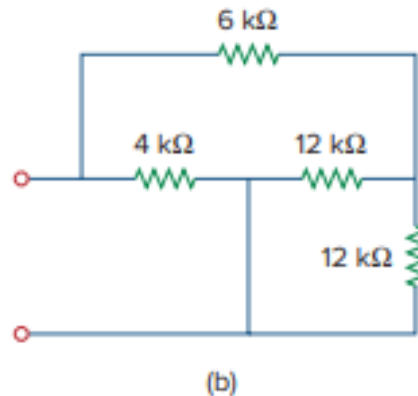
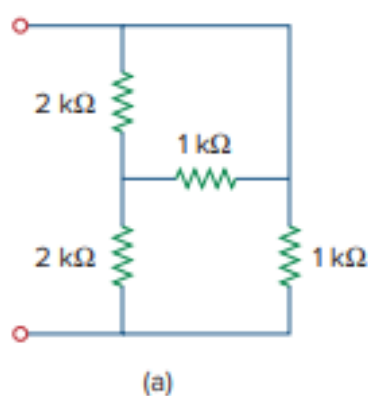


Tutorial

11. Using series/parallel resistance combination, find the equivalent resistance seen by the source in the circuit. Find the overall absorbed power by the resistor network.

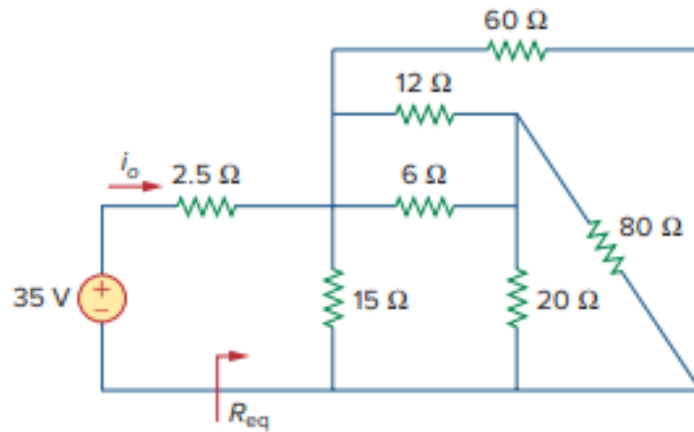


12. Evaluate R_{eq} looking into each set of terminals for each of the circuits.

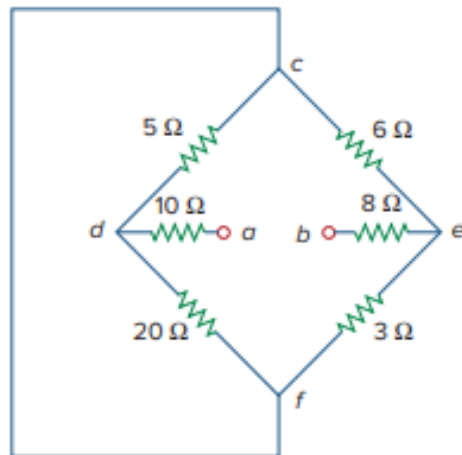


Tutorial

13. Find R_{eq} and i_o in the circuit.

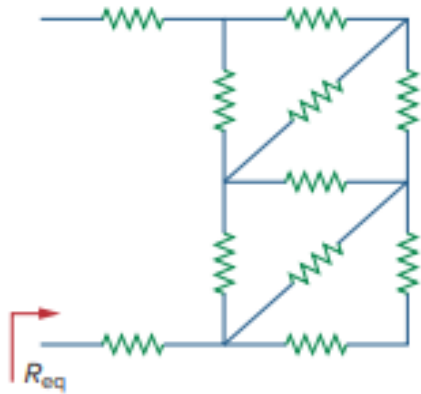


14. Find the equivalent resistance R_{ab} in the circuit.

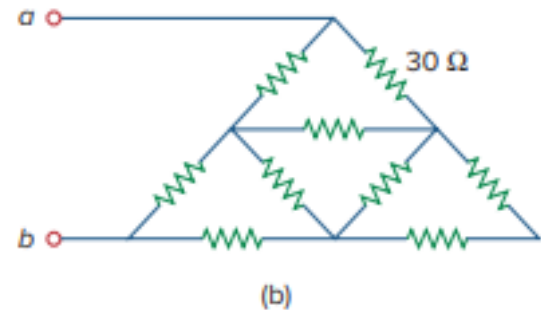
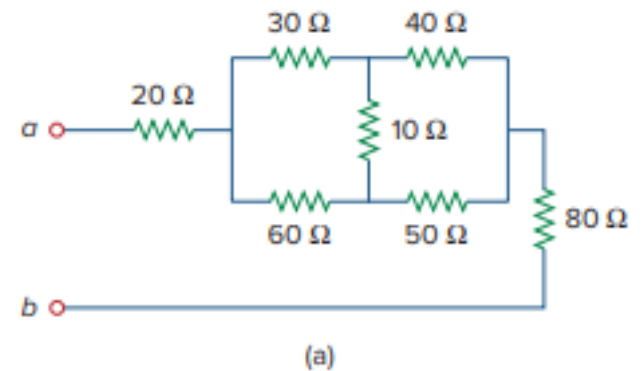


Tutorial

15. For the circuit shown below, find the equivalent resistance. All resistors are 3Ω .

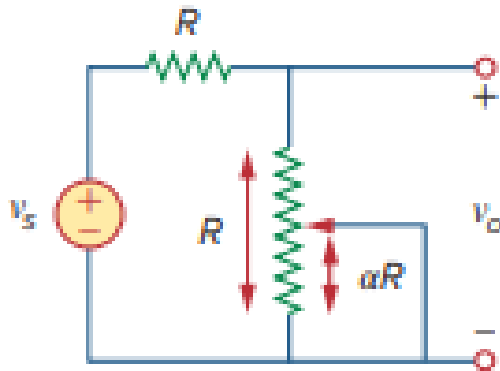


16. Obtain the equivalent resistance R_{ab} in each of the circuits (a) and (b). In (b), all resistors have a value of 30Ω .

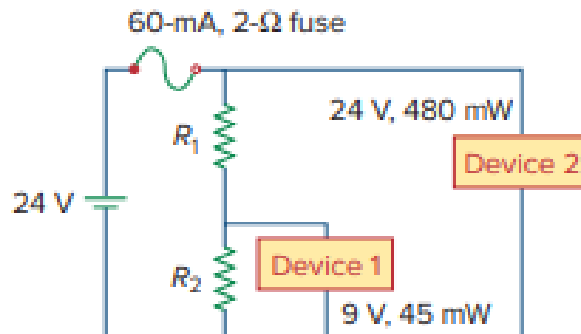


Tutorial

17. In the circuit, the wiper divides the potentiometer resistance between αR and $(1 - \alpha)R$, $0 \leq \alpha \leq 1$. Find v_o / v_s .



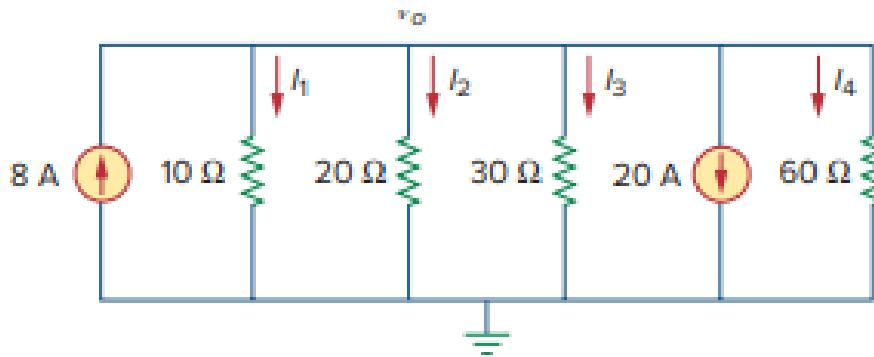
18. Two delicate devices are rated as shown. Find the values of the resistors R_1 and R_2 needed to power the devices using a 24-V battery.



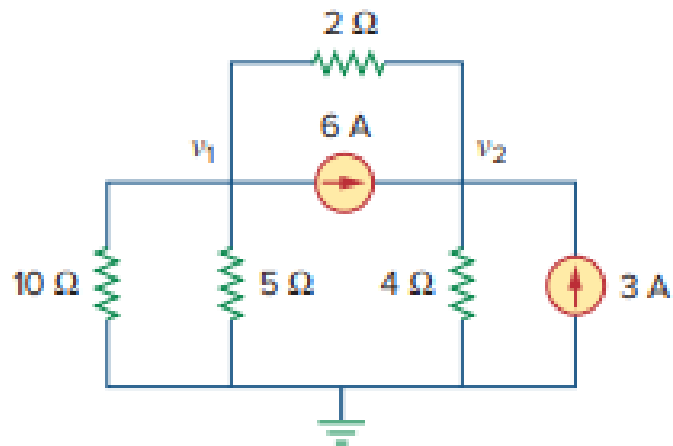
Node Analysis

Tutorial

1. Find the currents I_1 through I_4 and the voltage v_o in the circuit using Nodal analysis method.

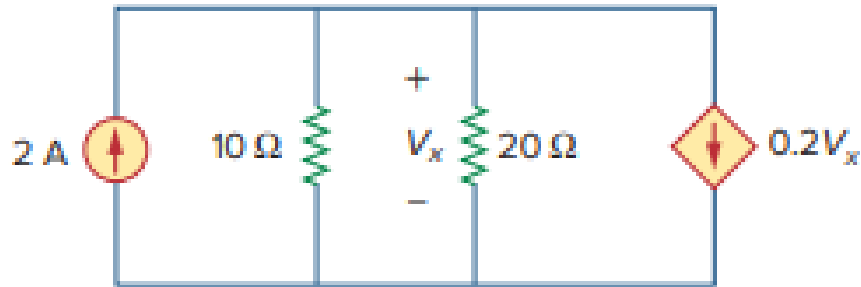


2. Find the voltages v_1 and v_2 using the nodal analysis method.

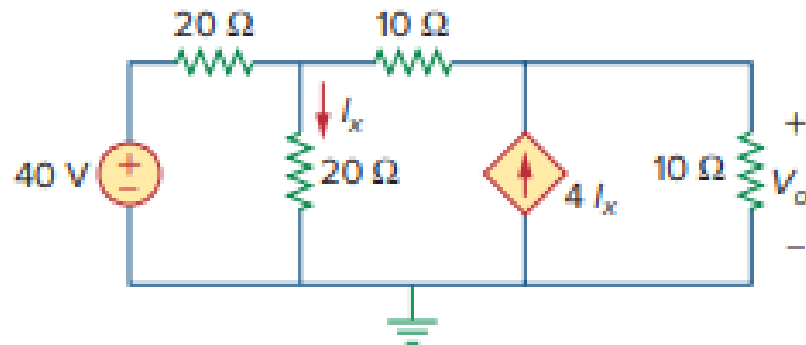


Tutorial

3. Apply nodal analysis to solve for V_x in the circuit.

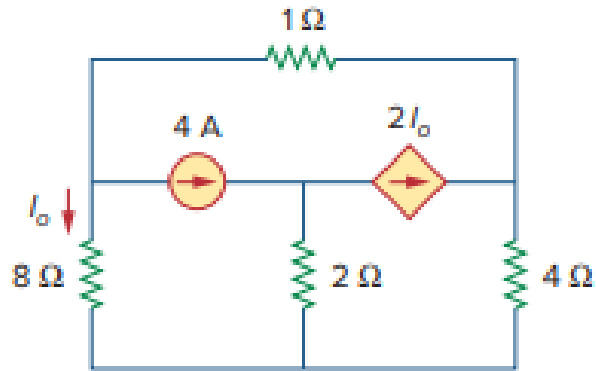


4. Using nodal analysis, determine V_o in the circuit.

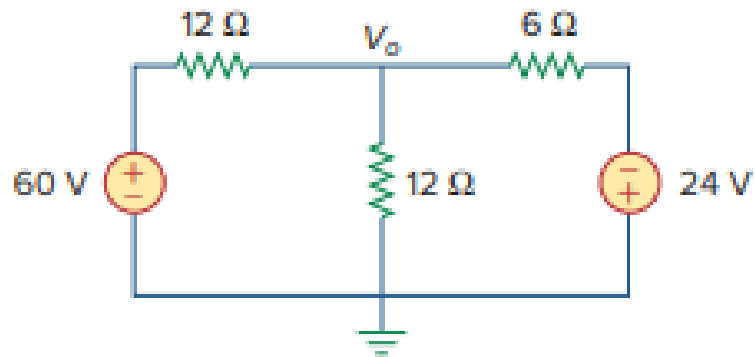


Tutorial

5. Find I_o in the circuit using Nodal analysis method.

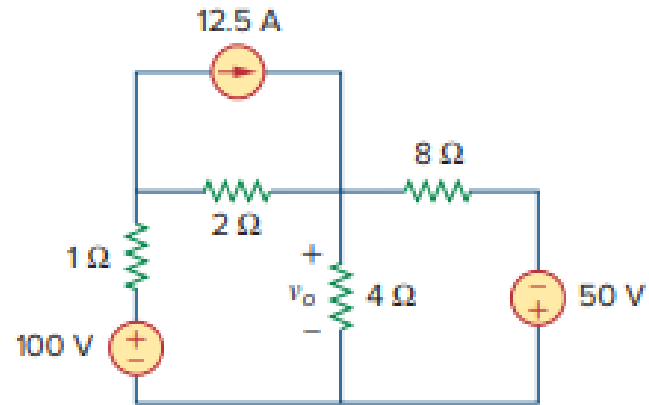


6. Find V_o and the power dissipated in all the resistors in the circuit. Use Nodal analysis method.

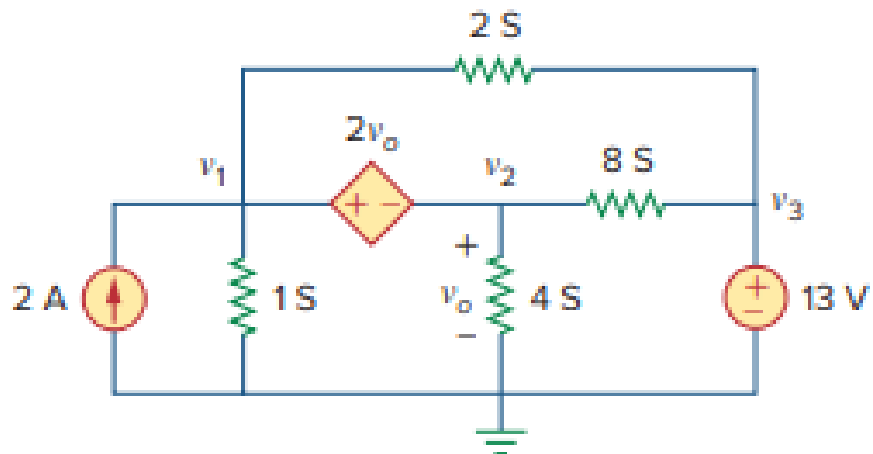


Tutorial

7. Using nodal analysis, find v_o in the circuit.



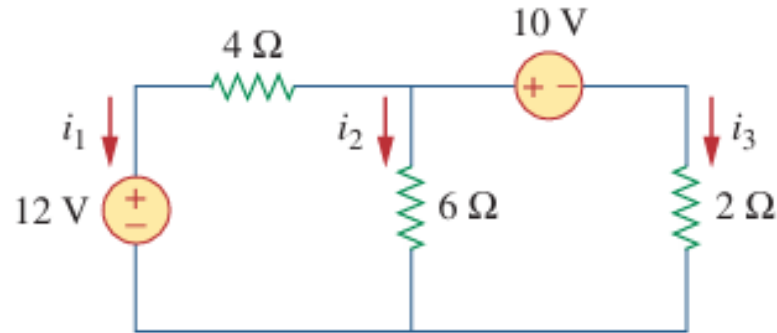
8. Determine voltages v_1 through v_3 in the circuit using nodal analysis.



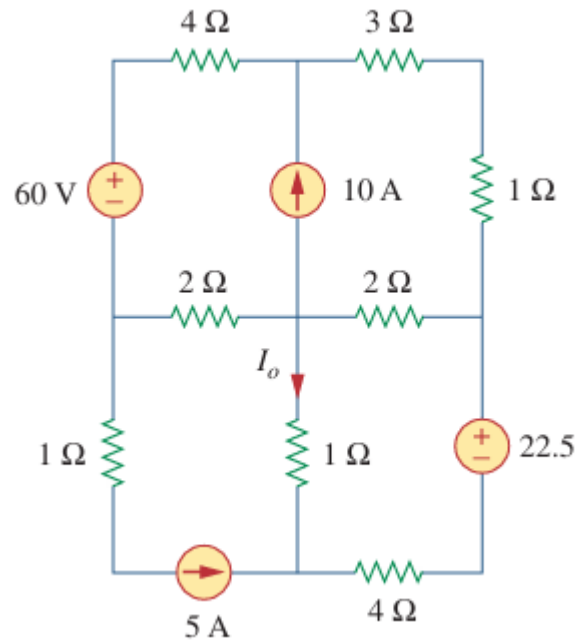
Mesh Analysis

Tutorial

1. Use mesh analysis to obtain i_1 , i_2 , and i_3 in the circuit

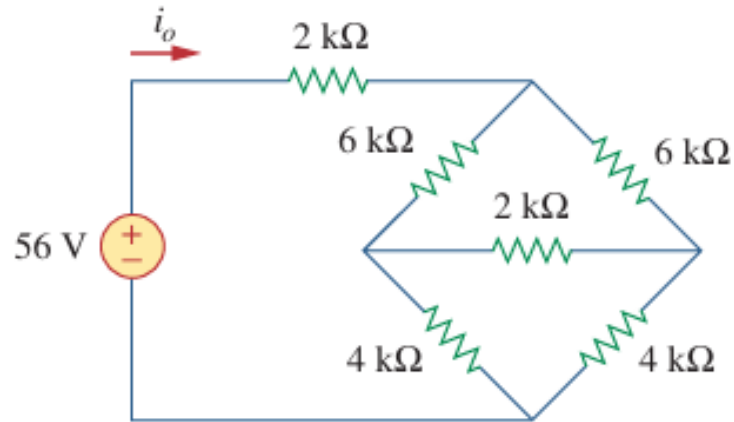


2. Apply mesh analysis to the circuit and obtain I_o .

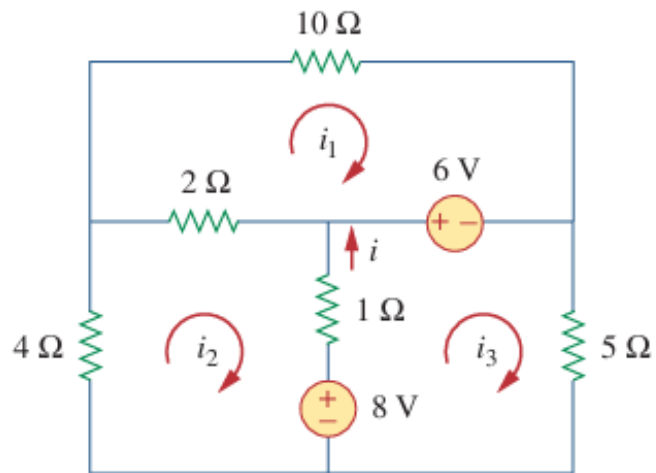


Tutorial

3. For the bridge network, find i_o using mesh analysis.

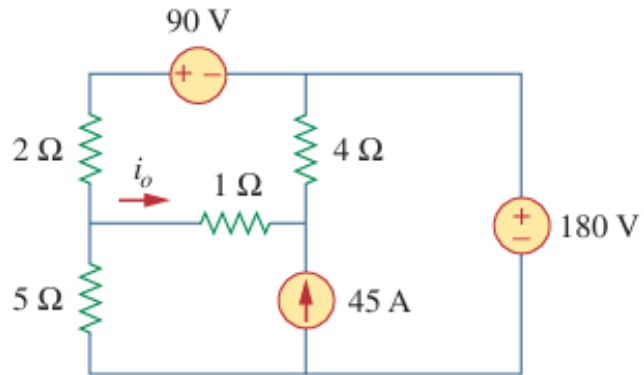


4. Apply mesh analysis to find i .

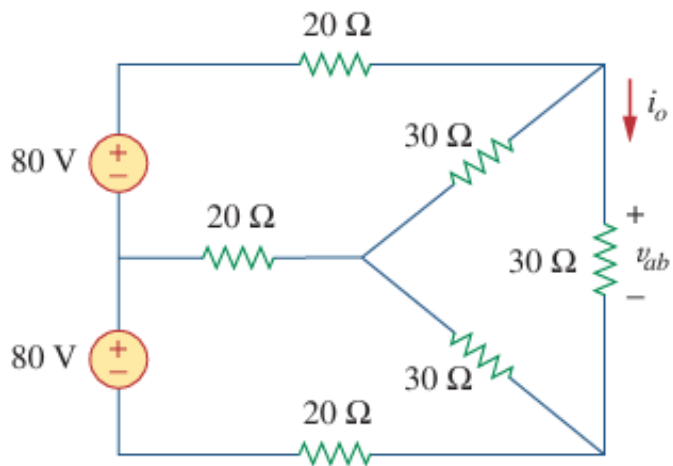


Tutorial

5. Use mesh analysis to obtain in the circuit.

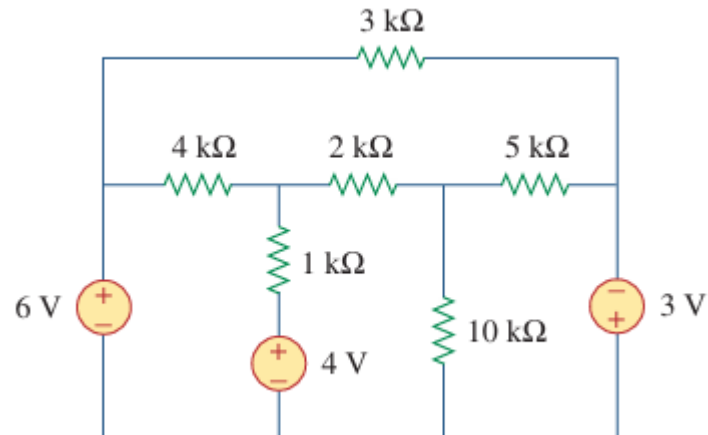


6. Use mesh analysis to find v_{ab} .

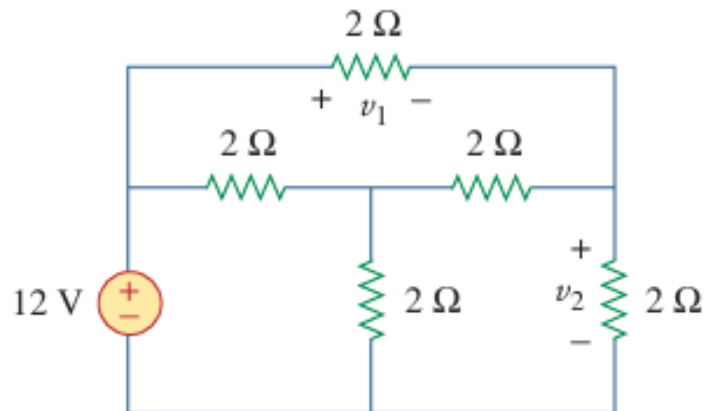


Tutorial

7. Determine the current through the 10-k resistor in the circuit using mesh analysis.

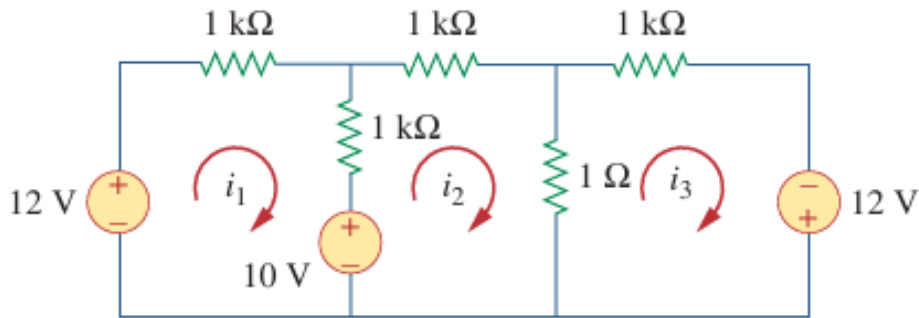


8. Determine voltages v_1 and v_2 in the circuit using mesh analysis.

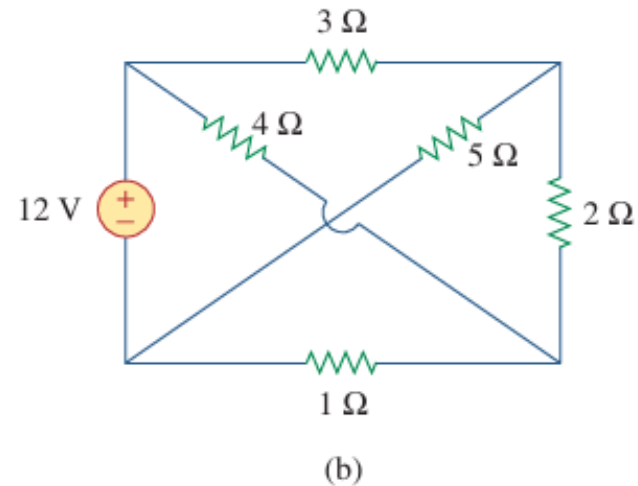
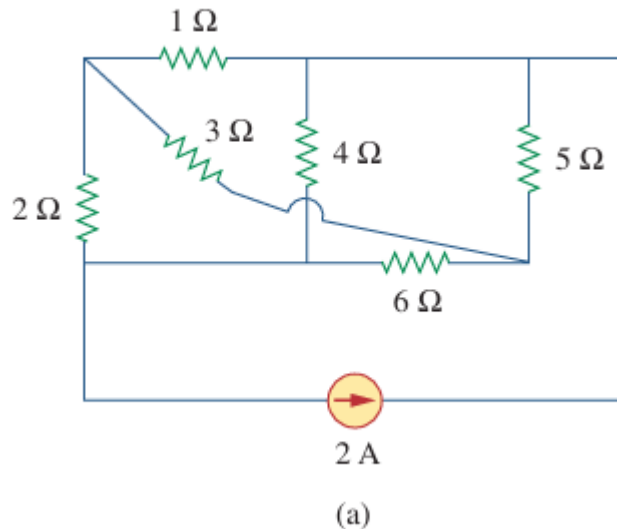


Tutorial

9. Find the mesh currents i_1 , i_2 , and i_3 in the circuit.



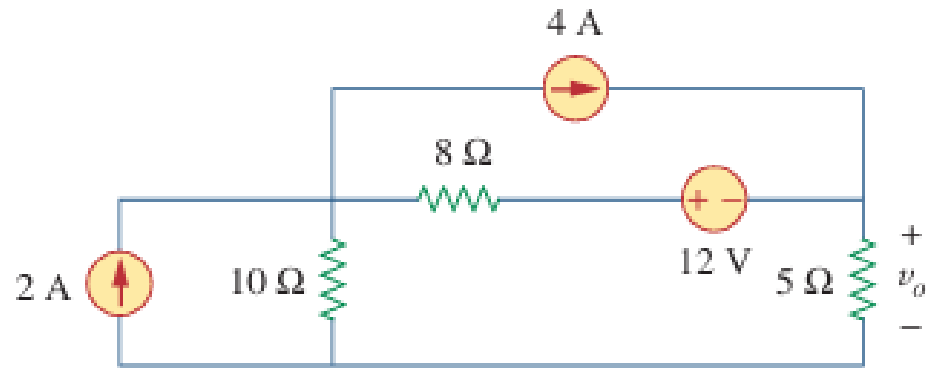
10. Which of the circuits is planar? For the planar circuit, redraw the circuits with no crossing branches.



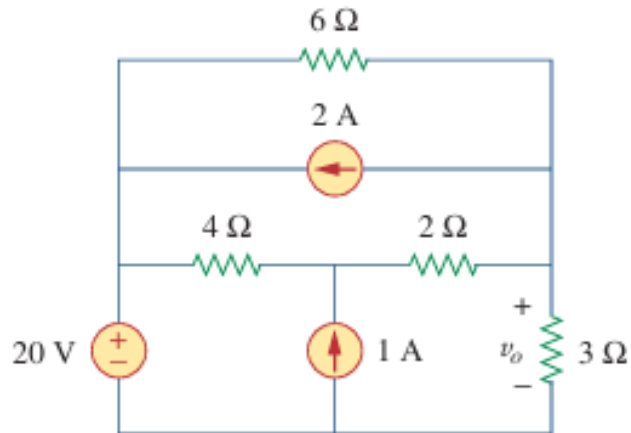
Thevenin's Norton's, Superposition, Maximum Power Transfer, Reciprocity Theorem

Tutorial

1. Using Superposition theorem find v_o .

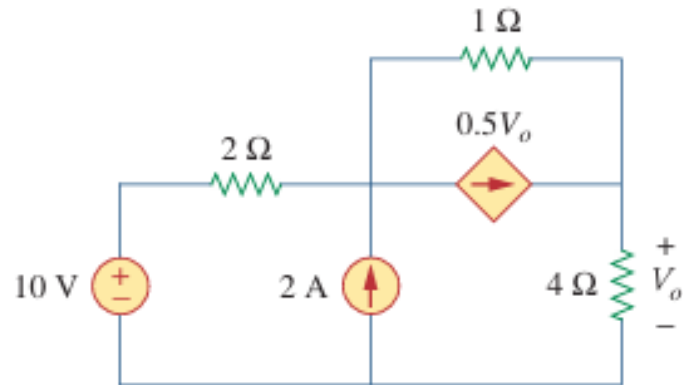


2. Apply the superposition principle to find v_o circuit .

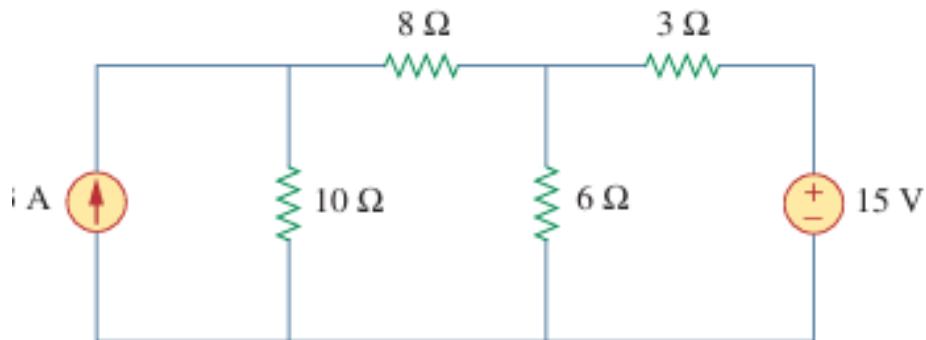


Tutorial

3. Use Superposition to find in the circuit.

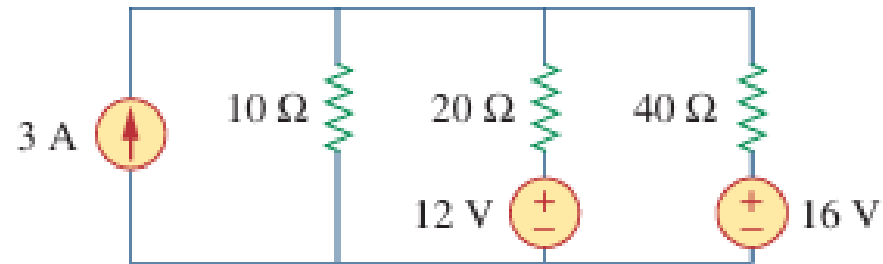


4. Referring to the circuit below, use source transformation to determine the current and power absorbed by the 8 Ω resistor..

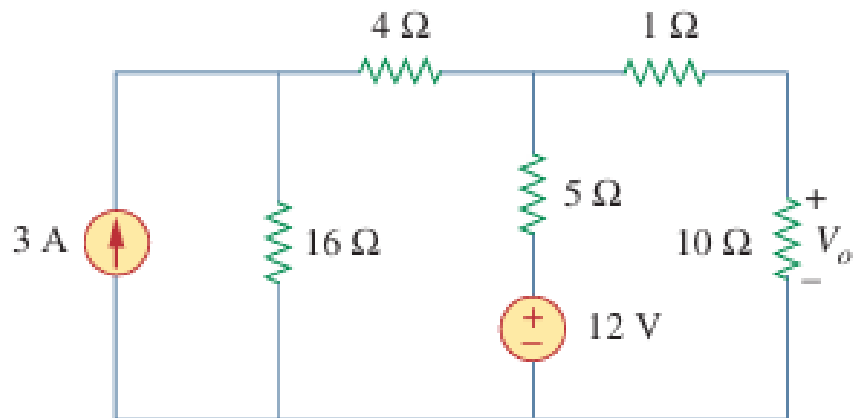


Tutorial

5. Use source transformation to reduce the circuit below into a single voltage source in series with a single resistor.

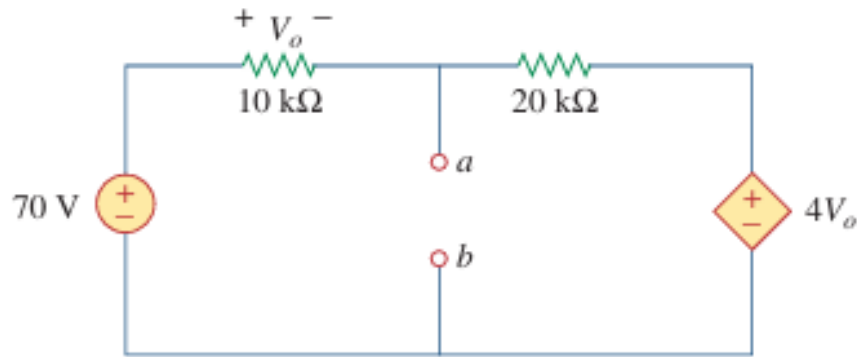


6. Apply Thevenin's theorem to find V_o in the circuit.

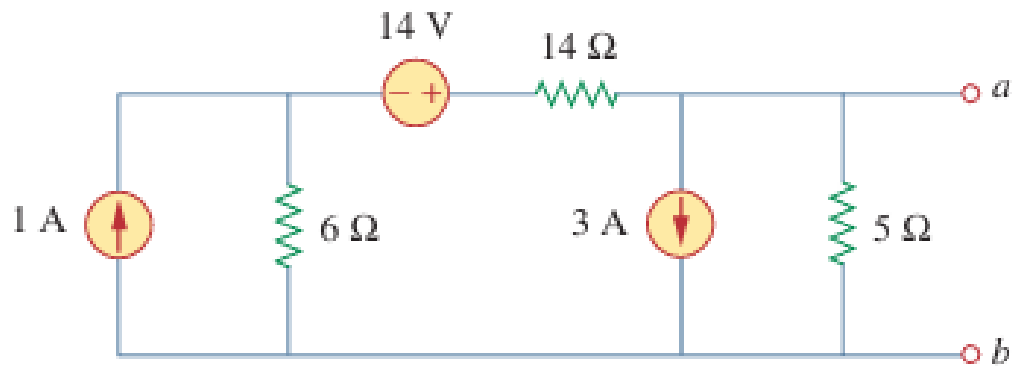


Tutorial

7. Find the Thevenin's equivalent at terminals a-b of the circuit.

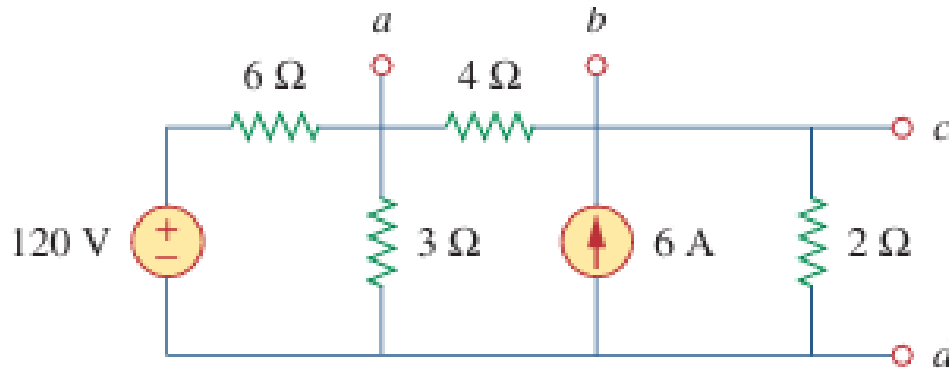


8. Find the Thevenin and Norton equivalents at terminals of the circuit.



Tutorial

9. Given the circuit, obtain the Norton equivalent as viewed from terminals i) a-b ii) c-d.



10. Compute the value of R that results in maximum power transfer to the $10\text{-}\Omega$ resistor in circuit. Find the maximum power.

